

Lab 1 - PECARN TBI Data, Stat 214, Spring 2026

2026-02-20

1 Introduction

The premise of this data analysis is to reduce the unnecessary use of Computed Tomography (CT) scans used on pediatric patients when attempting to diagnose clinically-important Traumatic Brain Injuries (ciTBI). This is building off of an analysis done by Kupperman et al. (2009) in *The Lancet*. I will note here that Kupperman et al. define a ciTBI as follows: clinically-important TBI is defined as a priori as death from TBI, neurosurgery, intubation (assisted breathing) for more than 24 hours for TBI, or hospital admission of two or more nights for the TBI in association with TBI on CT.

When a pediatric patient is presenting to an emergency department with a traumatic brain injury (TBI), the health-care specialist must make a difficult decision. They must weigh the risks of addressing the immediate threat to the child by performing a CT scan against the long-run risk to the future adult that comes with exposure to radiation at such a young age. An effect that disproportionately affects women over men.

The goal of Kupperman et al. was to identify a clinical decision rule (CDR) that can be used in the field to help make this decision. Specifically, to try and better predict when the CT scan could be obviated in patients at very low risk of ciTBI. CT scans are the predominant way of testing for TBI's. They are fast, don't require the patient be still for an extended period, and can see things that an MRI cannot. The number of CT scans has been on the rise, and the use of CT scans to diagnose TBI's in pediatric patients is higher in general emergency departments than in specifically pediatric emergency departments.

Better understanding the relationship between the features present in the dataset, and coming up with an interpretable model, would give health-care specialists the ability to more confidently obviate a CT scan in favor of the patient's long-term health. In the moment, when the current dangers feel overwhelming, it would be helpful to have an industry standard rule for making this decision. Especially in cases when the patient is pre-verbal and is not able to communicate. This report will attempt to present the data in an interpretable way, highlight some key findings, and rank the CDR implemented by Kupperman et al., logistic regression, and a model of our choice in terms of accuracy and usability.

2 Data

2.1 Data Description

The data we will be analysing is the result of a prospective cohort study of patients younger than 18 years of age with head trauma in 25 emergency departments of the Pediatric Emergency Care Applied Research Network (PECARN). Patients were enrolled in the study from June 2004 to March 2006. In total, there were 43,399 patients enrolled in the study and whose observations are present in the dataset. The dataset has 125 features, but not all features are present for each patient. Many features have a specific code to represent “not applicable,” which is different from the standard NaN that represents missing data.

Many of the features are dependent on each other, or can be grouped into various categories: See figure 1 on the next page. For example, if the patient was experiencing vomiting, there are additional variables that measure the intensity. such as, how many episodes of vomiting? When did it start? When was the last episode? There are also various measurements to summarize the patient’s responsiveness to stimuli, these are represented in the figure by the Glasgow Coma Scale (GCS) score. The scale is from 1-15, with lower numbers representing more severe head trauma. The scale is slightly different for patients that are under two years old. On that note, there are features that are not recorded if the patient is under two years old: amnesia, headache, and dizziness.

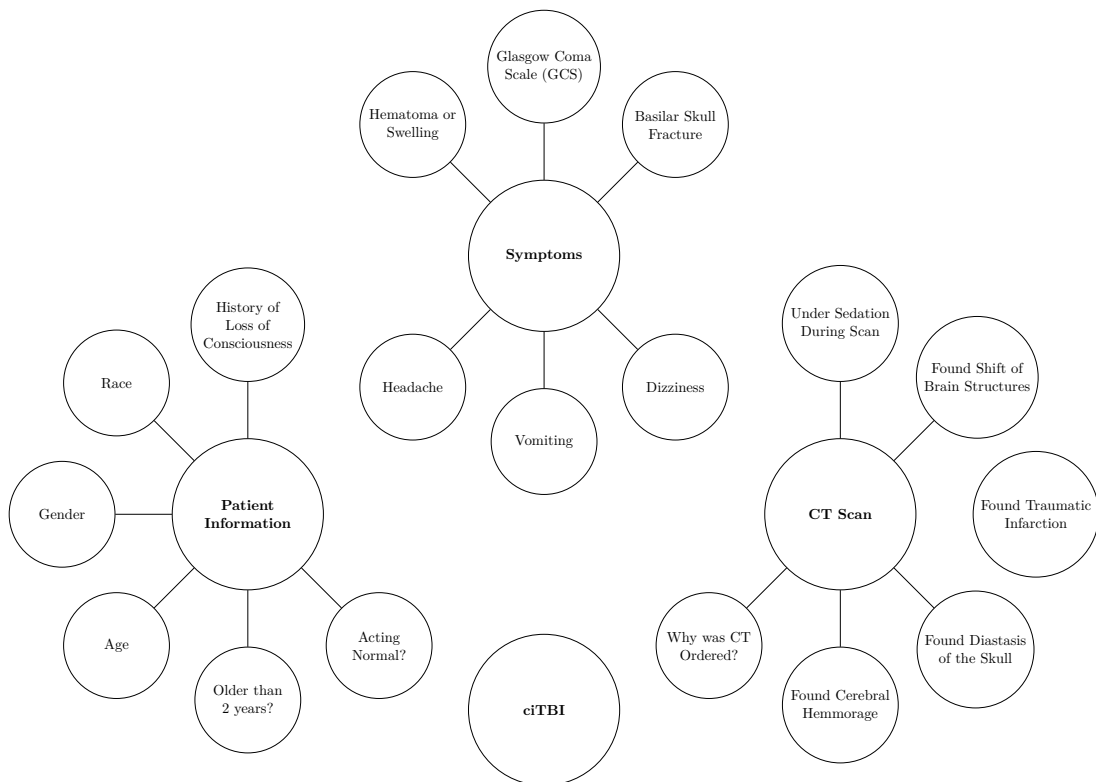


Figure 1: Examples of observations included in the dataset. (not complete picture)

This Data is exactly what we would need to construct our model to better predict when a CT scan is not needed. We have a lot of information regarding the pediatric patients that are brought into emergency rooms. It turns out that, by design, most of the dataset involves patients with a

GCS of 14-15, meaning they are responsive and aren't obviously presenting symptoms of a ciTBI. If we can use this dataset to generate an accurate model that predicts from these patients whether they actually do have a ciTBI, then we could potentially deploy the model and decrease the number of extraneous CT scans.

2.2 Data Collection

A good amount of the data was collected on site in the 25 emergency departments. Children presenting within 24 h of head trauma were eligible. Some exceptions include, children with trivial injury mechanisms, patients with known brain tumors, or patients that had already had neural imaging done at a different facility. Trained site investigators and other emergency department physicians recorded observations on a standardized data sheet. At times, a second assessment would be done within 60 minutes of the first. CT scans were interpreted by an on-site radiologist. To identify potentially missed traumatic brain injuries, research coordinators performed standardized telephone surveys of guardians of patients 7 to 90 days after the emergency department visit. If the guardian(s) could not be reached, available records were reviewed to ensure that no discharged patient was subsequently diagnosed with ciTBI. This included county morgue records.

All of the data has been encoded in numeric form. For example, **gender** is encoded as 1: **male**, 2: **female**, and **Finding 1** (cerebellar hemorrhage found during brain scan) is encoded as 0: **No**, 1: **Yes**, 92: **Not Applicable**. **Finding 1** would be listed as 92 if a patient was never given a CT scan. In general, the specific encodings for other options are: 92: **Not Applicable** is placed in a feature that is not used for that patient, 91: **Pre-verbal/Non-verbal** is the catch all code for the patient is not speaking, and 90: **Other** is used when describing something that does not fit into the features preset list of categories. For example, **EDDisposition** (the patient's next destination) is stored as the numbers 1 - 8, then 90: 1: **Home**, 2: **OR**, 3: **Admit - general inpatient**, etc. Now that the data collection process and the structure of the data has been explained, we can now look at the ways in which this data may not be in a state to do modelling just yet.

2.3 Data Cleaning

This data is not clean, and arguably is not "tidy" either. By including the 92 option, we are now distinguishing between not-applicable and NaN (missing data), but we already have a way of knowing if something was not applicable. e.g. why do we need 92 for **VomitNbr** (number of vomiting episodes since injury) in that variable to tell us that another variable (**Vomit**) was marked as 0: **No**. Choosing to encode the data in this way is a little confusing, and produces outliers in the dataset that will skew our model. Having many large values for **VomitNbr** actually skews the mean and artificially inflates the importance of this and other variables like it. We run into similar problems for the variables containing 90 and 91 as well.

We can also look more closely at the amount of missing/inconsistent data. In my training split, I was able to find 32,552 missing values overall. The two features with the most missing data being **Dizzy** (a yes or no variable) and **Ethnicity** (Hispanic or Non-Hispanic). Each of these can be assumed to mean that the site investigator could not get a definitive answer from either the patient or their guardian(s). See figure 2 for some specific details.

In some good news, I did not find any values that were mislabeled when they should have been 92. i.e. For all of the dependencies, when the required variable was 0, the extra variable was 92 (or missing) indicating it was not applicable. My final comment will be that the variable names

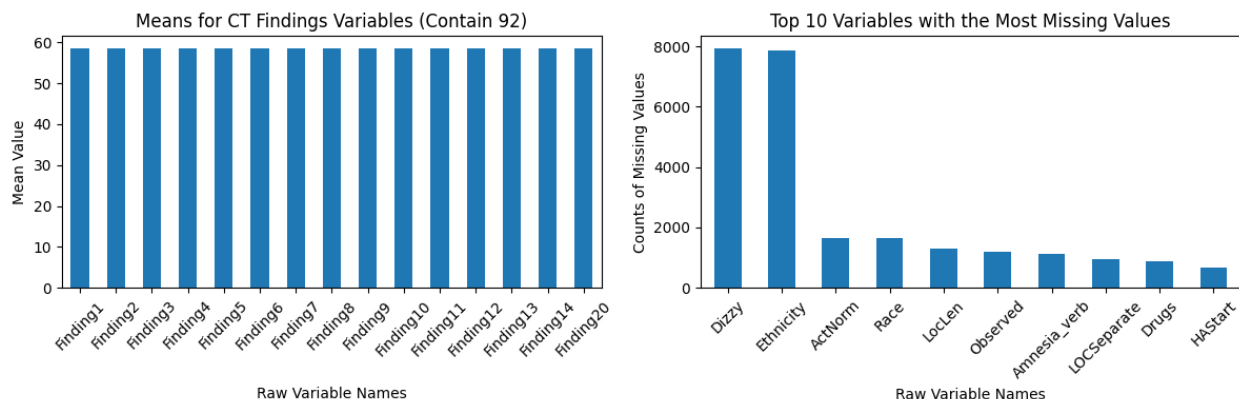


Figure 2: Consequences of Extreme/Missing Values in Training Dataset

themselves are inconsistent and not interpretable to someone with limited domain knowledge. Why is it written `HA_verb`, but a later variable is `HASStart` (no underscore). Similarly, I had to look up `EDDisposition` as it is assumed the reader of the data dictionary knows what that is.

2.4 Data Cleaning

The data cleaning process will follow these basic steps:

Step	Action	Justification
1	Remove rows with GCS less than 14 and missing <code>PosIntFinal</code>	This mirrors the Kupperman et al. article. We are only interested in becoming better at obviating CT scans for patients with little to no risk of ciTBI, so we will use the GCS score as our estimator for that risk. Second, the few datapoints that are missing the primary outcome will not be useful in our analysis as they do not have a response variable for us to compare our prediction to.
2	Create <code>verbal</code> indicator feature from <code>92</code> (0 is “pre-verbal/non-verbal”, 1 is “responsive”)	This specific encoding (<code>92</code>) held easily accessible information, and so I believe it should be its own variable. I believe that a domain expert would agree with this replacement.
3	Replace the encodings <code>90</code> , <code>91</code> , and <code>92</code> with <code>0</code> , <code>np.nan</code> , and <code>np.nan</code> respectively	This forces one method of missing or not-applicable data. <code>90</code> is replaced with <code>0</code> instead of <code>NaN</code> because the “Other” it represents category still contains some relevant information, we just want it to be on the same relative scale as the other levels in the categorical data. Additionally, <code>92</code> is now its own feature, which means that <code>92</code> now represents missing information.

Step	Action	Justification
4	Check that there are no inconsistencies between dependent features. For example, if <code>Vomit</code> is marked as 0, but <code>VomitNbr</code> is marked at 2	While I could not find any observations like this in my ad-hoc exploration, that does not mean that it is not present in the validation or test data (or real, future data). Therefore, we should protect against that inconsistency. Note that I am choosing to prioritize the global variable over the dependent if a disagreement arises.
5	Standardize column names to lowercase without underscores	Improves consistency across the dataset, makes analysis easier if all column names follow the same structure

Above are the explicit steps made in the data cleaning process. You can look at `clean.py` in the `code/` directory to see how I implemented those steps in code. However, there is one main implied judgement call that I must make clear. I have decided that it is acceptable for the cleaned dataset to contain missing values. The main reason for this is that when I explored the training data (which you can follow in `lab-1-explore.ipynb` in the `code/` directory) there were many missing values. Importantly, a key feature has many missing values: `Ethnicity`. Recall that `Ethnicity` is either 1: `Hispanic` or 2: `Non-Hispanic`.

I did not feel comfortable imputing this variable with the mode or median. Doing so would assign the missing values to the majority ethnicity, which is not reflective of reality. Additionally, forcefully erasing minority ethnicities is not very kosher. However, If I tried to use the mean, then I am imputing it with a non-interpretable value: something between 1 and 2. I could even try and make the portion of the patients with missing ethnicities match the portions in the rest of the data, but then I am assigning ethnicities randomly and is that really better than just not having it in the first place? Lastly, I did not feel comfortable removing all of the patients whose ethnicities are missing, because as you can see in figure 2 it's almost 8000 patients, which with a 50% train split is roughly 1/3 of the training data.

2.5 Data Exploration (On Clean Data)

Now that we have defined the data cleaning process, we can start looking at relationships among variables in the clean dataset. We can start by looking at the distributions of the patients.

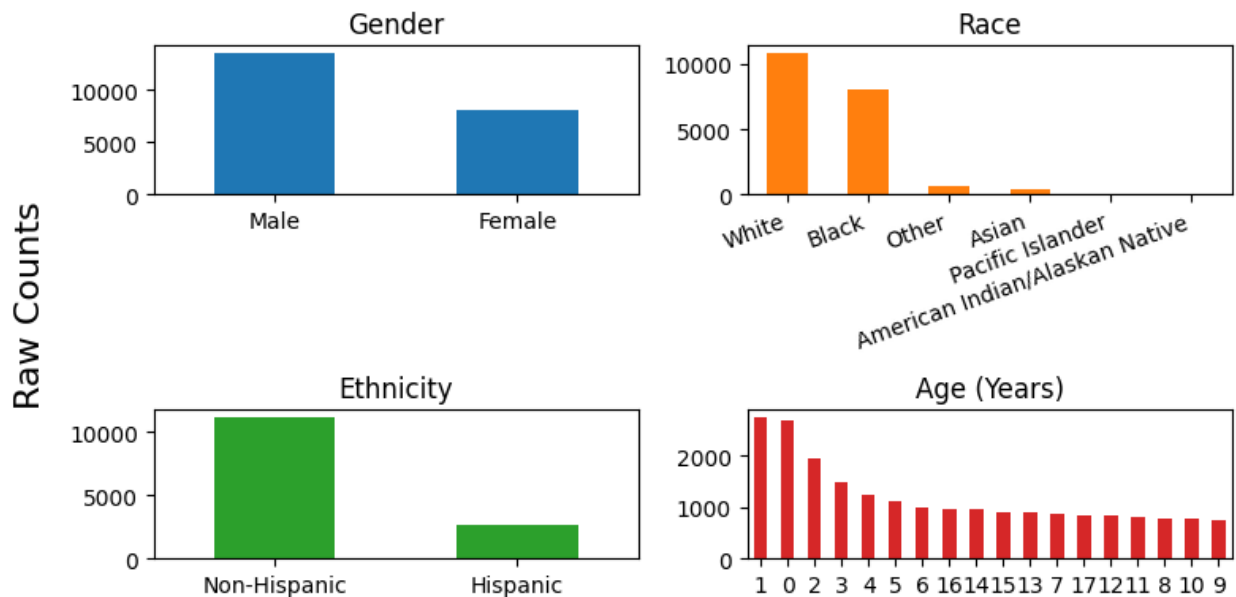


Figure 3: Counts of Patients by Gender, Race, Ethnicity, and Age

Figure 3 gives us a birds-eye view of the different types of patients that were presenting at these 25 emergency departments within 24 hours of a TBI. Predominantly Male, White, and young. However, if we look at the percentages of patients in these groups that receive a CT scan, we get a different story.

Somewhat surprisingly, in figure 4 on the next page there is an equal percentage of male and female patients that are receiving scans, similarly with Hispanic and Non-Hispanic patients. Although, we need to keep in mind that we are missing a lot of data when it comes to the ethnicity of patients. The high percentage of American Indian/Alaskan Native and Pacific Islander receiving CT scans can be attributed to their very low population count in the dataset, so it only takes a small number of CT scans to boost their percentage.

Lastly, in figure 4 it appears the trend in Age has reversed. In raw counts, the older you were, the fewer patients were presenting at the ED with TBI. However, the older the patient, the more likely they are to have received a CT scan. This may be due to the lower perceived risk of long-term effects on older patients. With the exception of the increase for *very* young patients. I imagine this is due to them being unable to communicate and the fact that they are more fragile, so a CT scan may be the best/fastest way to ascertain if there is immediate physical danger to the patient.

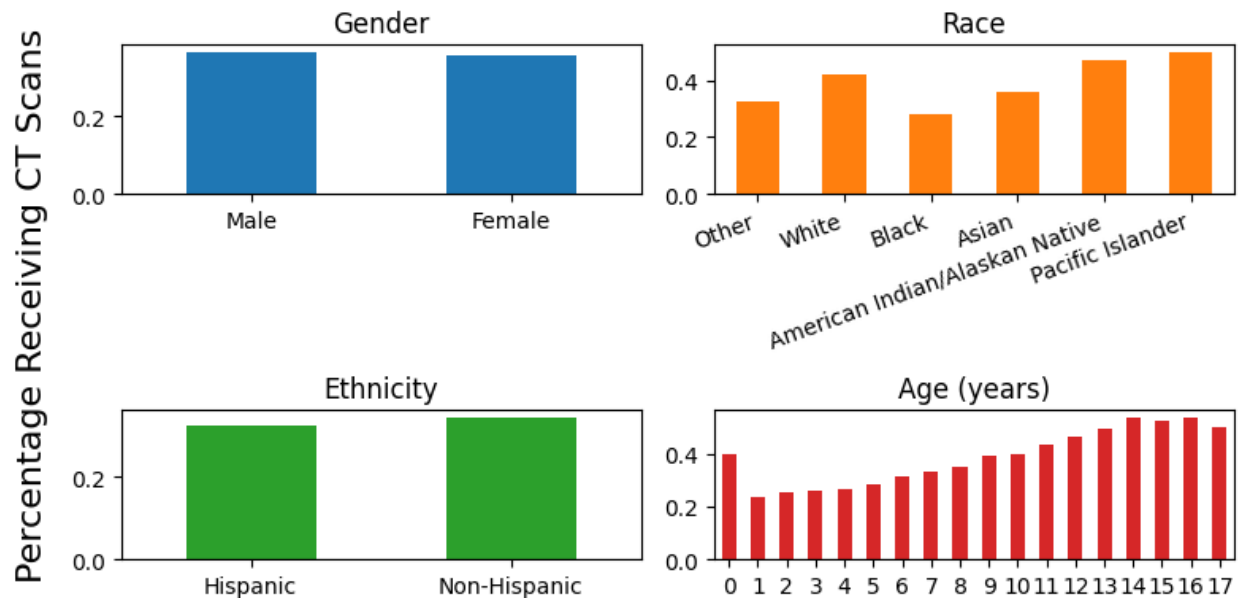


Figure 4: Percentages Receiving CT Scans for Gender, Race, Ethnicity, and Age

3 Findings

3.1 First finding

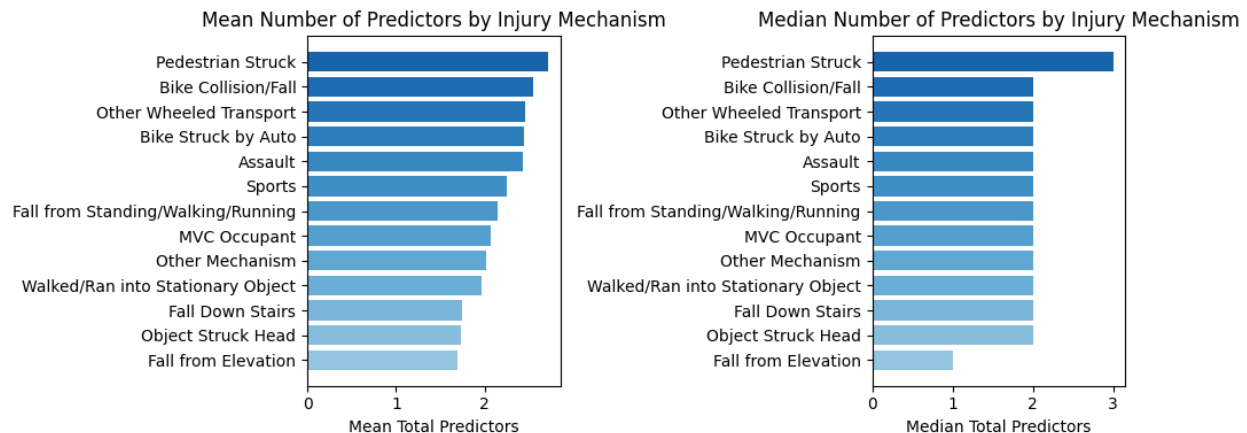


Figure 5: Mean and Median Number of Indicators Per Injury Mechanism

Figure 5 was created by aggregating over the different primary columns that many of the other data features were dependent over. Specifically, by counting the number of these variables each patient was marked as having, then calculating the mean and median total number per injury mechanism. It's not too surprising which ones lead to potentially more injuries. However, what is concerning is that the mean of **Fall from Elevation** is much higher than its median, and this implies there are some falls where patients are injured enough to present multiple predictors of TBI's. Even if their GCS was 14-15.

3.2 Second finding

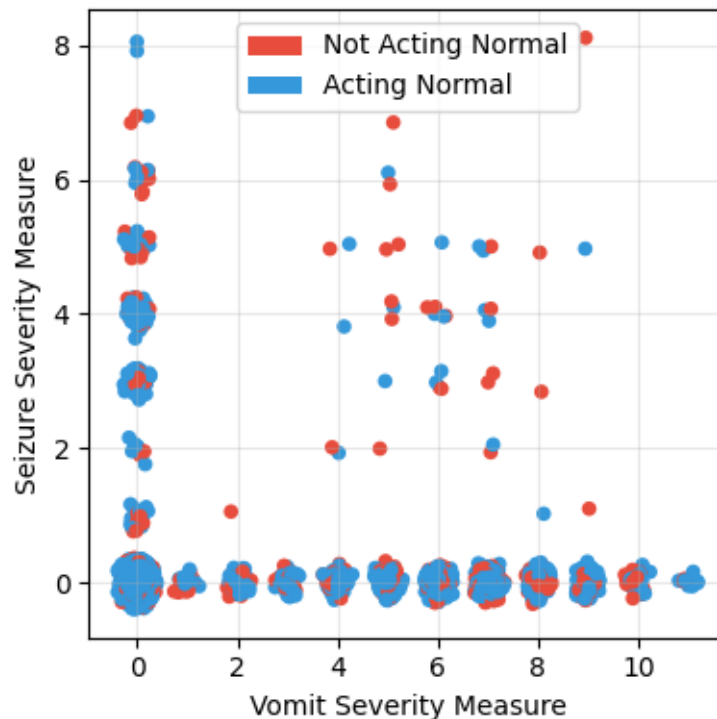


Figure 6: Seizure vs Vomit intensities with Gaussian Noise Grouped by Behavior

Figure 6 was generated by aggregating over the seizure predictors and the vomit predictors to generate a single measure for the intensity of those predictors. Then, some gaussian noise was added in both directions so that densities could be seen in the scatterplot. Now, while there is some comorbidity between these two measures. It is clear that they primarily appear independently. With seemingly almost all patients presenting one or the other at varying levels of intensity. There also does not appear to be a bias towards perceived behavior, as not natural clusters appear.

I'm afraid this was as far as I was able to get into the Lab. I do have the Academic integrity, collaboration, and bibliography sections immediately after this. I apologize for the inconvenience.

4 Academic Honesty

4.1 Statement

To Professor Yu,

I attest that the work in this report is my own and that all sources are properly cited, including my classmates. Academic research honesty is necessary for the validity of scientific results. Without academic research honesty, we lose the veridicality of the facts that arise as a result of scientific research. If one, as a researcher, is dishonest regarding data or its sources then the results of one's research is no longer relevant. If a researcher takes from another without credit, then their overall credibility as a researcher is now in question. These concerns are why I believe academic research honesty is important.

Sincerely,
your STAT 214 student.

4.2 Collaborators

I discussed the project briefly with fellow students Victoria Karai, Tobias Roemer, and Abhishek Shukla. I leaned on AI to create a starting point for the figures present in the report, and asked it questions to help with debugging or to remind me what commands I could use to achieve my specific data needs.

4.3 Bibliography

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